More Structural Shocks than Observables in DSGE Estimation: A Short Technical Note

1 State-Space Representation

A linear(ized) DSGE model can be written in state-space form:

$$x_t = F(\theta)\, x_{t-1} + G(\theta)\, \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q), \tag{1}$$

$$y_t = H(\theta)\, x_t + u_t, \qquad u_t \sim \mathcal{N}(0, R), \tag{2}$$

where x_t is a vector of latent states, y_t are observables, and ε_t are structural shocks. Shocks affect observables indirectly through the state vector and the model's propagation mechanism; thus counting arguments based on the number of shocks and observables are not, by themselves, decisive. See An and Schorfheide (2007), Herbst and Schorfheide (2015), Durbin and Koopman (2012).

2 The Kalman Filter and the Likelihood

The likelihood integrates out the latent state path:

$$\begin{aligned} p(y_{1:T} \mid \theta) &= \int p(y_{1:T}, x_{1:T} \mid \theta)\, dx_{1:T} \\ &= \int \left[\prod_{t=1}^{T} p(y_t \mid x_t, \theta) \right] p(x_{1:T} \mid \theta)\, dx_{1:T}. \end{aligned} \tag{3}$$

Under Gaussian assumptions, the Kalman filter yields the prediction-error decomposition:

$$\log L(\theta) = \sum_{t=1}^{T} \left[-\frac{1}{2} \log |S_t| - \frac{1}{2} \nu_t' S_t^{-1} \nu_t \right] + \text{constant}, \tag{4}$$

where $\nu_t = y_t - \mathbb{E}[y_t \mid y_{1:t-1}]$ is the one-step-ahead forecast error and $S_t = \mathrm{Var}(\nu_t \mid y_{1:t-1})$ its variance. This formulation makes clear that the likelihood is well-defined without a one-to-one mapping from shocks to observables; identification instead relies on the model's dynamic structure and cross-equation restrictions. See Durbin and Koopman (2012), Hamilton (1994), An and Schorfheide (2007).

Useful explainers. Wikipedia (Kalman filter): `https://en.wikipedia.org/wiki/Kalman_filter`. Standard technical references: Durbin and Koopman (2012) and Hamilton (1994). A DSGE-focused state-space primer is An and Schorfheide (2007).

3 Why Medium-Scale NK Models Can Sustain Many Shocks

In medium-scale NK models (e.g., Smets and Wouters 2007), different shocks enter different blocks of the model (Euler equations, Phillips curves, capital accumulation, policy rules) and propagate through common state dynamics. Identification typically derives from distinct persistence and distinct dynamic responses (IRFs) across observables.

4 When "Too Many Shocks" Becomes a Problem

Difficulties arise when shocks are observationally equivalent (similar entry points, similar persistence, similar propagation), leading to weak identification and "variance swapping." A separate concern is that shock flexibility may absorb structural misspecification and distort inference about structural parameters. See Den Haan and Drechsel (2021). Correlated or dynamically interacting disturbances can also matter empirically; see Cúrdia and Reis (2012).

5 Identification Criterion

The relevant condition is not:

$$\#\text{shocks} \leq \#\text{observables}. \tag{5}$$

Instead, identification depends on rank / local invertibility conditions: whether the mapping from structural parameters (and shock process parameters) into the likelihood-implied moments (or directly into the state-space likelihood) is locally one-to-one. See Iskrev (2010) for a formal treatment of local identification in DSGE models.

6 Post-2008 Financial-Friction Models

Post-2008 DSGE models commonly add financial wedges/shocks (risk premia, credit spreads, intermediation constraints, capital quality/valuation, liquidity, and sometimes housing/collateral constraints) to match spread dynamics and macro-financial comovement. This added flexibility can improve empirical fit but increases the risk of observational equivalence among shocks and of absorbing misspecification. Good practice includes reporting variance decompositions and IRFs, checking identification diagnostics, and testing shock specifications (cf. Den Haan and Drechsel 2021; Cúrdia and Reis 2012).

Representative references include Bernanke et al. (1999), Iacoviello (2005), Gerali et al. (2010), Gertler and Karadi (2011), Christiano et al. (2014).

Conclusion

More shocks than observables is not inherently problematic in DSGE estimation. The likelihood is computed by integrating over latent states, and identification comes from dynamic structure and cross-equation restrictions. Problems arise when shocks generate indistinguishable dynamics or absorb misspecification—risks that are especially salient in macro-financial models.

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